Robotics Report: A Golf Swing Simulation

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Implementation

The group implemented the movement by analyzing the angles between the joints and generated it using interpolation of keyframes. Discrete points in time for critical parts of the swing were defined: Backswing, Before Impact, After Impact and Follow-Through, associated with the respective joint angles. During the simulation, an interpolation function is used (`np.interp`), sampling the trajectory at each timestep and generating angular setpoints for MuJoCo's position actuators.

Simplifications

Rigid Lower-Body Anchoring: The pelvis and feet are rigidly constrained to the world frame. This eliminates leg kinematics and the weight-shifting dynamics typical of a human golf swing.

Constrained Spinal Motion: Abdomen mobility is restricted to rotation around Z and X; movement around the Y axis is locked.

Passive Wrist Mechanics: Wrist joint actuators remain undriven in the trajectory controller, omitting the biomechanical wrist cocking and uncocking phases that accelerate the clubhead.

Unactuated Left Arm (Passive Follower): Actuators for the left arm are disabled. The left arm acts as a passive kinematic follower linked to the right hand through a point-to-point constraint.

Rigid Club Attachment: The golf club is rigidly attached to the right hand coordinate frame, without shaft flexibility, grip compliance or hand-slip dynamics. The left hand is also connected to the right hand.

Open-Loop Trajectory Control: Joint targets are generated using time-indexed linear interpolation across five keyframe poses. The controller operates in open loop without dynamic feedback or contact impulse adaptation.

Additional Simplifications: The player and club use simplified geometric shapes. The left forearm and hand each have a mass of 0.0001 kg, reducing their dynamic contribution. Aerodynamic effects and explicit actuator torque limits are omitted, limiting the realism of ball flight and human effort.

Model Parameters

The simulation uses a timestep of 0.005 s and the `implicitfast` integrator. The club shaft is 0.58 m long, with shaft and head masses of 0.12 and 0.20 kg. The ball has a radius of 0.02135 m and a mass of 0.04593 kg. Initially supported by a tee, its motion results from gravity and contact forces, without a prescribed trajectory.

Disturbance Tests

The script `perturbations.py` executes a reference swing without external torque and separate disturbance tests on the right shoulder, elbow and wrist. Each simulation starts from identical initial conditions and lasts 6 s. Torques of -5, -1, $+1$ and $+5\,\mathrm{N\,m}$ are applied during $4.35 \leq t < 4.50$ s. The tested local axes are shoulder X, elbow $-Y$ and wrist X. Coverage is limited to these three axes.

Data Recording

CSV files record time, ball position, velocity, speed and individual actuator efforts, including their instantaneous maximum absolute value. Joint angles and external torque are not recorded. Results are saved under `src/tools/results/`, and `grafics.py` generates comparison plots under `src/tools/grafics/`.

Results and Discussion

The reference ball reaches a maximum speed of 11.38 m/s and a maximum centre height of 0.78 m, travelling predominantly along positive world Y. Shoulder disturbances of $+1$ and -1 N m produce maximum speeds of 11.63 and 9.06 m/s, respectively. An elbow disturbance of -5 N m reduces this value to 2.37 m/s.

Stored reference recordings match, and disturbed recordings coincide with them before torque application. Disturbance direction affects shot performance, and responses need not scale proportionally with torque because impact conditions change.

The model demonstrates sensitivity to disturbances through dynamics and contacts. However, its simplifications, limited axis coverage and the absence of comparison with real measurements limit physical validation.